

Rotating Equipment Incident Review

Incident IR-2025-017 | Pump P-204 drive-end bearing seizure | Approved 21 November 2025

Event overview

On 12 November 2025, process-water pump P-204 tripped following drive-end bearing seizure. No personnel were injured. Production was interrupted for 11.5 hours while the bearing, coupling, and damaged seal were replaced.

Evidence timeline

Time before trip	Evidence
21 days	Lubrication work order postponed due to production demand
8 days	Vibration increased above 7.0 mm/s RMS
2 days	Bearing temperature exceeded 86 C
Event	Metallic noise followed by bearing seizure and automatic trip

Root cause

The investigation identified lubrication starvation followed by progressive drive-end bearing degradation. The decisive warning was not any single measurement, but the compound pattern of postponed lubrication, sustained vibration above 7 mm/s, rising bearing temperature, and metallic noise.

Lessons learned

1. Treat simultaneous vibration and temperature excursions as a high-priority compound risk. 2. Do not postpone bearing lubrication beyond seven days. 3. Link work-order status with inspection trends so reliability teams receive one actionable warning. 4. Require maintenance clearance before restoring full load.

Applicability

The corrective actions apply to all P-100 and P-200 series pumps, including P-101 in Utilities Cooling Loop A.